

AMPV: Reproducible reliability mapping of astronomical photometric workflows under compound contamination

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Abstract

Aggregate benchmark scores can conceal where an astronomical measurement workflow becomes unreliable as observing conditions change. AMPV is a reusable experimental-design template for retaining that conditional structure: matched stochastic identities are propagated across explicit stress coordinates, workflow information classes are kept distinct, and continuous error and operational vulnerability are reported cell by cell rather than collapsed into a global ranking. The primary comet-like experiment evaluates workflows A–E on five reference-contrast levels and three contamination levels, using 300 matched identities to form 4,500 condition-specific scenes and 22,500 workflow evaluations. A separately frozen compact-PSF extension uses 300 identities and 18,000 evaluations of blind workflows A, B, C, and E. In the comet-like family, all 60 adjacent-reference-contrast paired comparisons had 95% bootstrap intervals below zero for the change in absolute error, whereas contamination effects were not uniform. In contaminated compact-PSF cells, all 32 adjacent-reference-contrast intervals were below zero and all 40 contamination contrasts were adverse. A further pre-outcome-frozen transfer test applied the same design logic to SEP 1.4.1 photometry outside the A–E measurement implementation: across 100 matched identities and 900 measurements, all six contamination-versus-clean contrasts

28 were interval-supported as adverse and all four contaminated-domain
29 reference-contrast comparisons were interval-supported as improving
30 with higher tested contrast. Workflow-relative behavior and vulner-
31 ability boundaries differed between the two controlled task families.
32 These results show the value of preserving matched, cell-resolved val-
33 idation structure and demonstrate implementation transfer without
34 implying a task-independent workflow ordering. The study remains
35 single-fidelity and synthetic; the reported boundaries are restricted to
36 the tested domains and do not constitute observational validation.

37 **Keywords:** astronomical photometry, validation, synthetic perturbations,
38 robustness, reliability mapping, reproducibility

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40 1 Introduction

41 1.1 Motivation

42 Validation based mainly on a pooled score or global workflow ordering can
43 hide the conditions under which an astronomical measurement pipeline changes
44 from stable to vulnerable. The methodological problem addressed here is
45 therefore not simply how to compare workflows, but how to retain the stress-
46 coordinate, matched-scene, and information-class structure needed to iden-
47 tify where a performance conclusion holds and where it ceases to hold.

48 Astronomical photometry and source-extraction workflows have been eval-
49 uated with controlled simulations, injected sources, and survey-level valida-
50 tion. Comparisons on simulated images can expose condition-dependent be-

51 havior, including degradation in crowded or overdense scenes [1]. Injection
 52 frameworks such as HSC SynPipe place synthetic sources into real survey
 53 images and have revealed blending-dependent photometric caveats [2]. Foun-
 54 dational tools such as SExtractor provide automated source detection, de-
 55 blending, measurement, and classification capabilities [3], while large survey
 56 releases such as Gaia EDR3 include dedicated photometric validation and
 57 systematic-quality assessment [4]. End-to-end simulators such as PhoSim
 58 provide another route for studying pipeline behavior and systematics, while
 59 still requiring comparison with real data to establish fidelity in observational
 60 use [5]. The AMPV study is designed around that gap in evaluation struc-
 61 ture: instead of treating validation as a single ranking problem, it maps
 62 workflow behavior across a frozen stress grid and retains the cell-level evi-
 63 dence needed to identify reliability boundaries.

64 Astronomical image simulation and injection-based validation also pro-
 65 vide mature precedents for testing image-analysis systems. GalSim provides
 66 a general-purpose, high-precision framework for rendering synthetic astro-
 67 nomical sources [8]. Survey pipelines have been characterized by insert-
 68 ing synthetic sources into observational images: HSC SynPipe propagates
 69 injected objects through the HSC processing chain [2], while Balrog uses

70 large-scale injections into DES single-epoch images to sample selection and
71 photometric biases and the survey transfer function [9]. For task-specific al-
72 gorithm evaluation, the Blending ToolKit combines simulated blended-galaxy
73 images, standardized algorithm interfaces, and common metrics to support
74 principled comparisons of detection and deblending methods [10]. Taken to-
75 gether, these precedents establish the value of controlled simulation, source
76 injection, survey-level validation, and standardized algorithm comparison.
77 The complementary question pursued here is how to structure a comparative
78 validation experiment when the desired output is not only a pooled perfor-
79 mance summary but the tested region in which a workflow-level conclusion
80 remains supported. AMPV is positioned alongside, rather than prior to,
81 these approaches. The specific methodological gap addressed here is the loss
82 of stress-coordinate, matched-scene, and information-class structure when
83 validation is reduced to pooled performance summaries.

84 Common-random-number designs, paired comparisons, bootstrap uncer-
85 tainty, and domain-bounded validation are established ideas [6, 7]. AMPV
86 does not claim these individual ingredients as new. It provides a reusable
87 experimental-design template that combines them for astronomical workflow
88 validation while retaining five relations in one analysis: the matched stochas-

89 tic scene, the changed stress coordinate, the workflow information class, the
90 cell-resolved continuous error, and the state of an operational vulnerability
91 rule.

92 In the present implementation, those relations are retained through a
93 matched compound-contamination stress grid with preserved stochastic iden-
94 tities, explicit separation of blind and privileged-information workflow classes,
95 and cell-resolved continuous-error and operational-vulnerability surfaces. Its
96 intended output is therefore a tested reliability domain with explicit bound-
97 aries, not a task-independent workflow ranking. The later SEP transfer ex-
98 periment tests whether this design logic remains informative when the mea-
99 surement endpoint is moved outside the internal A–E implementation.

100 The present experiment uses a controlled synthetic environment so that
101 ground truth, perturbation strength, and repeated scene realization can be
102 fixed explicitly. [5] This makes it possible to separate changes in work-
103 flow output from changes in the underlying stochastic realization. Repro-
104 ducible provenance is retained as a supporting control; the scientific test is
105 whether preserving this matched structure exposes reliability boundaries and
106 workflow-relative changes that a pooled summary does not retain.

107 **1.2 From aggregate comparison to reliability mapping**

108 AMPV is framed here as a validation methodology rather than as an ad-
109 ditional photometric estimator. The objects being evaluated are workflows
110 A–E; AMPV supplies the experimental design, matched-scene structure, and
111 boundary-oriented interpretation used to assess them.

112 The V2 design crosses five reference-contrast levels with three contam-
113 ination levels and evaluates every workflow on the same matched scenes.
114 This structure supports two complementary views of performance: continu-
115 ous error behavior and a threshold-based vulnerability indicator. Together,
116 those views describe where error remains small, where vulnerability becomes
117 nonzero or increases, and how those patterns depend on the tested stress
118 coordinates.

119 **1.3 Relationship to CAPS**

120 AMPV and CAPS serve different roles in the present work. AMPV is the
121 validation methodology used here to organize stress testing, reliability map-
122 ping, and boundary analysis. CAPS is a controlled synthetic implementation
123 environment that can support such experiments. The V2 production used
124 in this manuscript was independently implemented within the AMPV V2

125 project; earlier CAPS V1 production outputs are not used in the present
126 analysis.

127 This distinction is important because the scientific contribution of the
128 present study is not a renamed workflow or a simple continuation of an
129 earlier ranking exercise. The emphasis is the reproducible methodology for
130 characterizing reliability across controlled perturbation space.

131 **1.4 Study objectives**

132 The present study has five objectives:

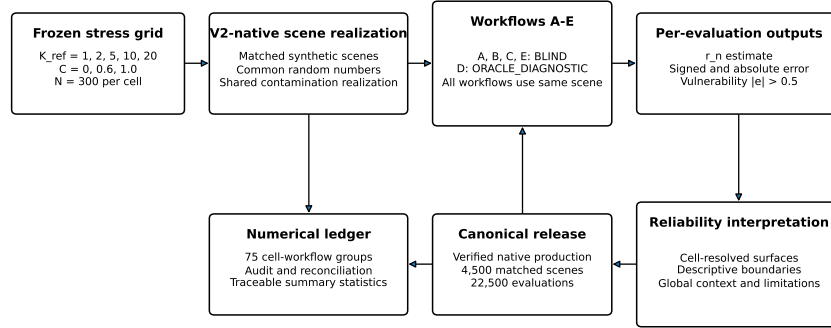
- 133 1. to define and execute a matched-scene stress grid over reference contrast
134 and contamination level for workflows A–E in the primary comet-like
135 family, with a separately versioned blind-workflow A/B/C/E extension
136 in the compact-PSF family;
- 137 2. to quantify both continuous error behavior and threshold-defined vul-
138 nerability at the cell level rather than relying only on pooled summaries;
- 139 3. to identify descriptive reliability boundaries while preserving workflow
140 information-class distinctions, including the diagnostic role of privileged-
141 information workflow D; and
- 142 4. to maintain a reproducible analysis chain linking reported numerical

143 results to the corresponding V2 production outputs; and
144 5. to test implementation transfer by applying the same matched reliability-
145 mapping logic to an independently implemented SEP photometry end-
146 point under a separately frozen compact-PSF subset.

147 The present study is single-fidelity and contains two separately versioned
148 controlled synthetic task families. The primary comet-like experiment is re-
149 tained unchanged. A compact-PSF extension specified and frozen before
150 outcome generation tests whether the AMPV reliability- mapping structure
151 remains informative after a material change in source morphology and esti-
152 mand. A later, separately frozen SEP transfer test addresses implementation
153 dependence without changing the underlying synthetic task family. Multi-
154 fidelity and observational dependence remain outside the evidence supported
155 by the current study.

156 **1.5 Scope of external context**

157 External context is limited to the literature cited here. Prior-art statements
158 are restricted to controlled simulation and source injection, photometric val-
159 idation, source-extraction comparison, common-random-number methodol-
160 ogy, and validation-domain framing supported by Refs. [1, 2, 3, 4, 5, 6, 7, 8,



AMPV = validation methodology; CAPS = one controlled synthetic implementation environment.
Current canonical experiment: single fidelity (F0). Reliability statements are restricted to the frozen tested grid.

Figure 1: AMPV validation framework. Matched stochastic identities are propagated across explicit stress coordinates while workflow information classes are retained; cell-resolved continuous error and an optional operational vulnerability rule define the reported reliability landscape.

9, 10, 11]. Claims about the present AMPV results remain grounded in the
versioned V2 analysis outputs.

2 Methods

2.1 AMPV validation design and statistical unit

AMPV separates the validation design from the particular photometric work-
flows being tested. In this study the design consists of (i) a truth-known task
family, (ii) explicit stress coordinates, (iii) stochastic identities preserved

168 across those coordinates, (iv) workflow information classes, (v) a continu-
169 ous error measure, and (vi) an optional operational vulnerability rule. The
170 specific source model, stress variables, and workflows can therefore change
171 between validation tasks without changing this design logic. The first con-
172 trolled implementation maps workflow reliability under compound contami-
173 nation and contains 300 base stochastic sample identities. Each identity is
174 propagated through 15 condition cells formed by five reference-contrast levels,
175 $\kappa_{\text{ref}} = 1, 2, 5, 10, 20$, crossed with three contamination levels, $C = 0, 0.6, 1.0$.
176 This produces 4,500 condition-specific scene instances. Each scene instance
177 is evaluated by workflows A–E, giving 22,500 workflow evaluations and 75
178 cell–workflow groups with 300 evaluations per group. The task family is
179 `COMET_LIKE_PRIMARY` at fidelity level F0.

180 Reference contrast and contamination amplitude are the two stress coor-
181 dinates in this controlled implementation: the former changes signal-to-noise
182 difficulty through source gain, while the latter scales structured contamina-
183 tion. The five contrast levels and three contamination levels were fixed as the
184 discrete stress grid used to span low-to-high controlled difficulty; they are ex-
185 perimental coordinates, not claimed population frequencies or survey-derived
186 cut points. The study therefore interprets transitions only at the tested cells

187 and does not assign physical significance to the numerical spacing between
 188 levels. They define the tested validation domain rather than implying that
 189 astronomical observing conditions are intrinsically two-dimensional.

190 The same sample identity preserves r_n , interference seeds, and the un-
 191 scaled structured-contamination realization across all 15 cells. Thus the
 192 4,500 condition-specific instances are not treated as 4,500 statistically in-
 193 dependent base scenes. This common-random-number design permits paired
 194 comparisons across stress conditions and workflows while reducing variation
 195 due solely to unrelated stochastic draws.

196 **2.2 Synthetic signal and compound contamination**

197 The implemented comet-like source is generated on a 256×256 pixel image.
 198 The central nucleus is represented by a normalized Gaussian component with
 199 $\sigma = 1.5$ pixels and is combined with a radial coma component proportional to
 200 r_n^2/ρ^2 after the implemented central capping and normalization. The canoni-
 201 cal benchmark uses a nucleus fraction of 0.3 and a coma fraction of 0.7. This
 202 construction is an intentionally controlled synthetic task and is not presented
 203 as a complete physical radiative-transfer model.

204 Compound interference contains six implemented families: I1, sky gra-

205 dient/spatial mottling/background galaxies; I2, a dust jet and anti-solar
 206 tail; I3, sparse cosmic-ray-like events; I4, field-star Gaussian PSFs; I5, low-
 207 frequency detector baseline drift; and I6, Gaussian readout noise. The struc-
 208 tured I1–I5 realization is shared for a given sample identity and scaled by C ,
 209 whereas I6 is retained separately. The realized r_n values in the production
 210 span 0.310041–4.988671.

211 **2.3 Reference-contrast calibration**

212 Reference contrast is controlled through a source-gain calibration. For the
 213 reference regime, $r_n = 1$ km and the clean source flux is measured in a
 214 30-pixel aperture. The reference noise denominator is the I6 readout-noise
 215 standard deviation multiplied by the square root of the aperture pixel count.
 216 For each target κ_{ref} , source gain is selected so that the reference clean-
 217 aperture flux attains the requested contrast. The resulting gains for $\kappa_{\text{ref}} =$
 218 1, 2, 5, 10, 20 are 13.9532646896, 27.9065293792, 69.766323448, 139.532646896,
 219 and 279.065293792, respectively. The same gain is used across the three con-
 220 tamination levels at a given target contrast. The production also records the
 221 realized sample-specific contrast separately from the configured κ_{ref} .

2.4 Photometric workflows and information constraints

Workflow A is a blind 30-pixel-radius aperture sum with no background subtraction. Workflow B is blind aperture photometry using a 30-pixel circular aperture and a fixed 50–90 pixel annulus; the annular mean is used for background subtraction. Workflow C is blind aperture photometry with the same aperture and background ring but uses a deterministic RANSAC-style background estimator (seed 20260814, 200 iterations, sample size 30, and an inlier threshold of 0.2, σ of the background pixels).

Workflow D is an `ORACLE_DIAGNOSTIC`, not an information-equivalent blind estimator. It uses the true r_n to select an adaptive background annulus with inner radius $\max[\lceil 11r_n \rceil, 35]$ pixels, subject to image-geometry limits, a 40-pixel annulus width, and median background estimation. It is therefore interpreted as a privileged-information diagnostic reference.

Workflow E is blind model fitting with a circular two-dimensional Gaussian plus constant background on a local image patch. The least-squares parameters are source center, amplitude, Gaussian width, and background; an analytic Jacobian is used, the Gaussian width is bounded to 1–20 pixels, and the optimization limit is 120 function evaluations. Fitted flux is $2\pi|A|\sigma^2$.

Each workflow has a separate noiseless calibration curve based on 40 log-spaced r_n values from 0.1 to 10 km. Measured flux is mapped back to r_n by log-flux interpolation. Calibration is constructed separately for each workflow and source-gain level.

2.5 Compact-PSF extension

A second controlled family, `COMPACT_PSF_PRIMARY`, was specified and frozen before outcome generation, then executed as a separately versioned extension after the comet-like production had been frozen. It contains 300 base identities crossed with the same five reference-contrast targets and three contamination levels. Because no scientifically meaningful privileged-information analogue of workflow D was identified before production, the extension evaluates blind workflows A, B, C, and E, giving 18,000 workflow evaluations.

The source is a centered circular Gaussian PSF with independently sampled width $\sigma_{\text{PSF}} = 1.5\text{--}4.0$ pixels and integrated truth flux F_{true} . A two-dimensional Latin-hypercube design (seed 20261003) samples PSF width and a narrow intrinsic-amplitude factor of 0.8–1.2. The comet-specific dust-jet/tail interference I2 is not reused. It is replaced, before outcome generation, by I2P: a task-neutral asymmetric elliptical local contaminant with

259 geometry and amplitude fixed relative to a reference signal scale and indepen-
260 dent of sample F_{true} and σ_{PSF} . The remaining stress construction preserves
261 matched sample identities across conditions.

262 For this family, workflows A–C retain the same aperture/background def-
263 initions, whereas workflow E fits the circular Gaussian plus constant back-
264 ground. The estimand is integrated flux, so relative error is $(F^{\text{est}} - F_{\text{true}})/F_{\text{true}}$.
265 A clean-domain audit showed only numerical-precision-level error for A–C
266 and E, so no post-hoc calibration surface was introduced. The produc-
267 tion and statistical analysis were frozen before scientific interpretation; the
268 latter used 2,000 sample-identity bootstrap replicates with seed 20261004.
269 The same primary $\tau = 0.5$ vulnerability definition and the same threshold-
270 sensitivity grid were retained.

271 **2.6 Independent SEP implementation-transfer exper-** 272 **iment**

273 To test whether the AMPV experimental-design logic depends on the photo-
274 metric implementation used in the A–E workflow set, a separately versioned
275 implementation-transfer experiment was frozen before outcome generation.
276 It reuses the compact-PSF scene construction but replaces the measurement

277 endpoint with SEP version 1.4.1 [11], an implementation outside the A-E
 278 workflow code and derived from the Source Extractor code base [3]. One hun-
 279 dred matched identities were evaluated at $\kappa_{\text{ref}} = 1, 5, 20$ and $C = 0, 0.6, 1.0$,
 280 giving 900 measurements. For every cell, SEP `sum_circle` circular-aperture
 281 photometry used the known source center, a 30-pixel radius, a 50–90 pixel lo-
 282 cal background annulus, and `subpix=0`. No per-pixel error, variance, or gain
 283 was supplied to SEP; its returned flux-uncertainty field is therefore not used
 284 as a scientific uncertainty estimate. Statistical uncertainty in this experiment
 285 is instead quantified by the sample-identity bootstrap described below. No
 286 source detection or recentering step was added, because the transfer ques-
 287 tion concerns the measurement implementation under the same truth-known
 288 matched scenes rather than detection completeness.

289 The integrated injected Gaussian flux remains the truth quantity. Signed
 290 relative error, absolute relative error, and the frozen $\tau = 0.5$ vulnerability rule
 291 are therefore unchanged. The transfer analysis uses 2,000 sample-identity
 292 bootstrap replicates with fixed seed 20261004. Before the N=100 produc-
 293 tion, a separate N=3 software smoke test was restricted to finite-output,
 294 row-count, and SEP-flag checks and was not used as scientific evidence or to
 295 tune the frozen design. The transfer criterion did not require SEP to outper-

form any internal workflow: support required a non-degenerate cell-resolved
reliability surface and bootstrap-supported systematic change along at least
one frozen stress axis. This experiment is an external-implementation trans-
fer test, not observational or survey validation.

2.7 Error quantities and operational vulnerability cri- terion

Signed relative error is defined with the family-specific estimand. For the
comet-like primary family (`COMET_LIKE_PRIMARY`),

$$e = (r_n^{\text{est}} - r_n^{\text{true}}) / r_n^{\text{true}},$$

whereas for the compact-PSF primary family (`COMPACT_PSF_PRIMARY`),

$$e = (F^{\text{est}} - F_{\text{true}}) / F_{\text{true}}.$$

In both families, absolute error is $|e|$. Continuous error is the primary mea-
sure of estimation behavior. The frozen operational vulnerability indica-
tor uses $|e| > \tau$ with the primary frozen operational threshold $\tau = 0.5$,

308 corresponding to a 50% relative-error cutoff. The value is an operational
309 demonstration threshold chosen for the controlled validation experiment,
310 not a survey requirement, physical transition, or universal astronomical ac-
311 ceptance standard. Accordingly, continuous error remains the primary evi-
312 dence, and the robustness of the derived threshold summaries is examined
313 at $\tau = 0.25, 0.4, 0.5, 0.6, 0.75$, and 1.0.

314 **2.8 Paired uncertainty and robustness analyses**

315 The production size of 300 identities per primary task family was fixed before
316 outcome generation as the replication budget for the cell-resolved matched
317 experiment. It is not presented as a power-optimal or universally sufficient as-
318 tronomical sample size. Because the scientific comparisons are paired across
319 stress cells, precision is evaluated from the observed sample-identity boot-
320 strap intervals rather than justified retrospectively from the realized effect
321 sizes.

322 Uncertainty analyses use the 300 sample identities within each task family
323 as the resampling units. A cluster-bootstrap replicate samples these identities
324 with replacement and carries each selected identity’s complete cross-condition
325 and cross-workflow block within that family, preserving the matched-design

dependence. The comet-like analysis used 2,000 replicates with fixed seed 20261002; the separately frozen compact-PSF analysis used 2,000 replicates with fixed seed 20261004. These analyses provide 95% intervals for cell-level median absolute error, median signed error, vulnerability, and paired contrasts. In the comet-like family, blind-workflow contrasts are evaluated within matched $(\kappa_{\text{ref}}, C, \text{sampleid})$ instances and workflow D is kept separate because of its oracle information class. The compact-PSF extension contains only blind workflows A, B, C, and E.

Workflow E has complete optimization diagnostic records for all 4,500 evaluations. The optimizer reports **success=True** for 4,497 evaluations and **success=False** with status 0 for three. The production validity rule classified computational failure on exception rather than optimizer status alone. A sensitivity analysis therefore recomputed affected summaries after excluding these three cases without altering their canonical validity labels.

2.9 Reproducibility and provenance

For each task family, the experimental design, seed namespace, workflow definitions, and stress grid were frozen before that family's production. The original comet-like analysis was not modified when the separately versioned

compact-PSF extension was added. Numerical summaries and figures were generated from the corresponding versioned production outputs without altering either experiment. These controls support reproducibility and guard against outcome-dependent redefinition; they are not treated as the scientific novelty of AMPV. Detailed provenance records are retained in the reproducibility archive rather than repeated in the scientific narrative.

2.10 AI-assisted workflow

OpenAI ChatGPT was used for English-language editing and manuscript organization and as a technical assistant for parts of Python, PowerShell, and LaTeX workflow development, review, testing, and reproducibility-audit organization. Scientific results were derived from frozen, version-controlled computational outputs and independently checked by the author. All study-design, methodological, interpretive, and submission decisions remained the author’s responsibility, and the author reviewed the final code and manuscript and takes responsibility for the work.

359 **2.11 Ethical applicability**

360 The present work is a computational validation study based on synthetic as-
361 tronomical scenes and simulation outputs. It contains no human-participant,
362 animal-subject, clinical-trial, or personally identifiable data. Institutional
363 human/animal research approval and informed consent are therefore not ap-
364 plicable to the current study scope.

365 **3 Results**

366 The primary comet-like analysis contains 300 base stochastic sample identi-
367 ties propagated through 15 condition cells, yielding 4,500 condition-specific
368 scene instances and 22,500 workflow evaluations organized into 75 cell-workflow
369 groups. Its grid comprises five reference-contrast levels (reference contrast =
370 1, 2, 5, 10, 20), three contamination levels ($C = 0, 0.6, 1.0$), and work-
371 flows A–E. The comet-like values reported in the following subsections are
372 derived from the corresponding versioned production and analysis outputs;
373 the compact-PSF extension is reported separately from its independently
374 frozen production.

3.1 Global descriptive behavior across the frozen grid

Across the full frozen grid, the workflow-level summaries were:

- Workflow A: $N=4500$, median absolute error=0.019, IQR=0.0051–0.073, median signed error=0.012, vulnerability rate=5.5%.
- Workflow B: $N=4500$, median absolute error=0.015, IQR=0.0040–0.060, median signed error=0.0057, vulnerability rate=4.8%.
- Workflow C: $N=4500$, median absolute error=0.061, IQR=0.016–0.232, median signed error=8.852e-04, vulnerability rate=15.3%.
- Workflow D: $N=4500$, median absolute error=0.015, IQR=0.0038–0.058, median signed error=0.0025, vulnerability rate=5.0%.
- Workflow E: $N=4500$, median absolute error=0.0034, IQR=8.988e-04–0.013, median signed error=8.213e-04, vulnerability rate=1.8%.

These aggregate summaries pool all 15 experimental cells and are used only as a compact descriptive overview. Cell-resolved behavior is the primary basis for discussing reliability boundaries. The pooled workflow-level summaries are shown in Fig. 2.

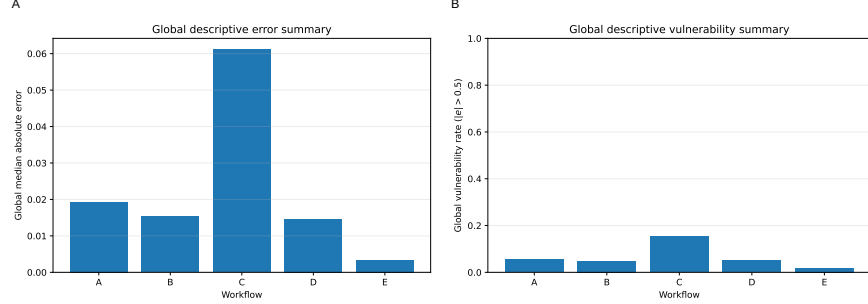


Figure 2: Global pooled descriptive context for the primary comet-like experiment. Values aggregate all 15 stress cells for orientation only; inferential and boundary claims in the text use matched, cell-resolved analyses rather than this pooled view.

3.2 Cell-resolved error and vulnerability landscape

The 15 experimental cells show substantial variation in median absolute error and in the fraction of evaluations exceeding the frozen operational vulnerability threshold ($|e| > 0.5$). The cell-resolved median absolute-error landscape is shown in Fig. 3, and the corresponding vulnerability landscape is shown in Fig. 4. For each cell, the observed numerical span is summarized by the surfaces rather than repeated cell by cell in the main text. Across the tested grid, the lowest cell-level median absolute errors occur for workflow E, whereas the largest values are associated with workflow C in the low-reference-contrast regime. Vulnerability shows the same broad concentration toward the low-contrast region, while the detailed workflow-relative structure changes with

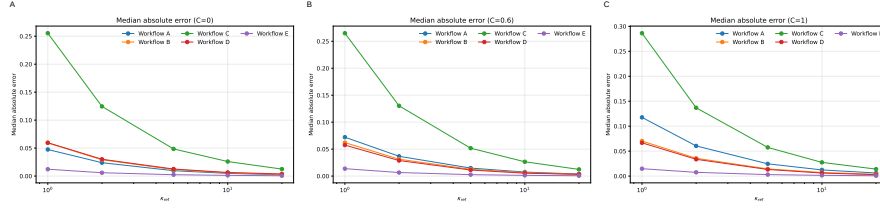


Figure 3: Cell-resolved median absolute-error landscape for the primary comet-like experiment across the tested reference-contrast and contamination grid. The surface is descriptive at the tested cells; paired sample-identity bootstrap contrasts are reported separately and no interpolation beyond the grid is implied.

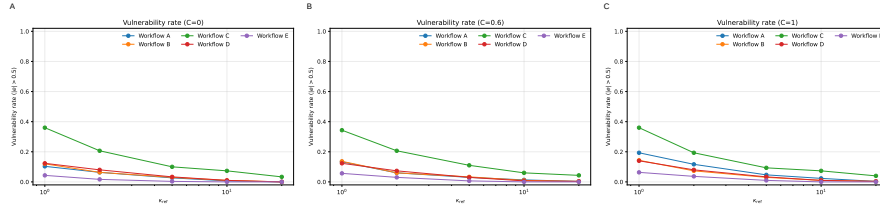


Figure 4: Cell-resolved vulnerability landscape for the primary comet-like experiment at the frozen operational threshold $|e| > 0.5$. The threshold is an operational summary rather than a physical transition; sensitivity to alternative thresholds is evaluated separately.

402 contamination. Complete values for all 75 cell–workflow groups are retained
403 in Table S1.

404 **3.3 Signed-error behavior**

405 Median signed error is retained separately from absolute error to distinguish
406 error magnitude from direction. Figure S1 shows the frozen signed-error
407 curves across the three contamination levels. Complete cell-level signed-
408 error values are included in Table S1. Directional structure is interpreted
409 only together with error magnitude, workflow information class, and the
410 corresponding stress-grid cell.

411 **3.4 Reliability-boundary framing**

412 The frozen grid is reported as a reliability landscape rather than as a single
413 global ranking. The cell-resolved summaries show that error magnitude and
414 vulnerability vary with both reference contrast and contamination level $\mathbf{C}$,
415 and that the magnitude of separation among workflows is not constant across
416 the grid.

417 At this stage, the data support reporting: - where median absolute errors
418 remain small or become large; - where vulnerability rates become nonzero

419 or increase; - how those patterns shift across reference contrast and C; - how
420 signed-error behavior differs from absolute-error behavior.

421 Interpretive explanations for why those patterns occur are reserved for
422 the Discussion.

423 **3.5 Paired uncertainty and robustness**

424 Cluster-bootstrap uncertainty was evaluated with the 300 base sample iden-
425 tities as the resampling units, preserving each identity’s complete cross-
426 condition and cross-workflow block. The resulting intervals are reported
427 with the derived-analysis tables. Paired blind-workflow contrasts confirm
428 that the separations visible in the cell-resolved landscape are condition de-
429 pendent rather than reducible to a single pooled ordering. In particular,
430 the paired absolute-error difference between workflows C and E is positive
431 in all 15 tested cells, with the 95% cluster-bootstrap interval above zero in
432 every cell. The corresponding A–E and B–E differences are also positive in
433 all 15 cells, whereas A–B changes with contamination condition and some
434 low-contrast intervals include zero.

435 Paired stress-axis contrasts sharpen this condition-dependent interpreta-
436 tion for the comet-like family. For contamination, 35 of 50 higher-*C* versus

437 $C = 0$ contrasts had 95% sample-identity bootstrap intervals entirely above
 438 zero, 13 included zero, and two were below zero; both negative intervals
 439 occurred for privileged-information workflow D at $\kappa_{\text{ref}} = 20$. Thus contam-
 440 ination predominantly increased absolute error but not uniformly across all
 441 workflow-contrast cells. In contrast, all 60 adjacent- κ paired contrasts had
 442 intervals entirely below zero for $\Delta|e| = |e|_{\kappa_{\text{high}}} - |e|_{\kappa_{\text{low}}}$, indicating lower
 443 absolute error at each successive tested reference-contrast level across all
 444 workflows and contamination settings.

445 Threshold sensitivity confirms that the numerical location of a threshold-
 446 defined boundary depends on the operational cutoff. The primary $\tau = 0.5$
 447 analysis exactly reproduces the frozen operational boundary table below. At
 448 $\tau = 0.4$, the A boundary at $C = 1$ moves from $\kappa_{\text{ref}} = 5$ to 10; at $\tau = 0.6$,
 449 the E boundary at $C = 0.6$ moves from 2 to 1 while the A-D primary
 450 boundaries are unchanged. More stringent or permissive cutoffs produce
 451 larger shifts. Continuous error surfaces are therefore treated as primary
 452 evidence and threshold boundaries as an operational summary.

453 Excluding the three Workflow-E evaluations with optimizer `success=False`
 454 changes the affected cell median absolute errors by at most 9.02×10^{-5} , in-
 455 dicating no material effect on the reported reliability landscape.

3.6 Compact-PSF extension

The second family contains 300 base identities and 18,000 workflow evaluations across 60 condition–workflow groups. All 40 paired contamination contrasts comparing $C = 0.6$ or 1 with $C = 0$ had 95% bootstrap intervals above zero. Of 48 adjacent- κ contrasts, 32 had intervals below zero; the remaining 16 were all $C = 0$ clean cells, where error was already at the numerical floor. Restricting the comparison to contaminated cells therefore gives 32 of 32 adjacent- κ intervals below zero (Table S3).

Workflow-relative structure also differed from the comet-like family. Within contaminated compact-PSF cells, workflow E had lower paired absolute error than A, B, and C throughout the tested grid; A had lower paired error than B and C, whereas B–C intervals included zero throughout those cells. The E result is interpreted as a family-specific model-match effect because the source truth is itself a circular Gaussian, not as evidence of universal workflow superiority. Threshold sensitivity placed operational vulnerability mainly in the low-contrast, high-contamination region; by $\kappa_{\text{ref}} \geq 5$, vulnerability was essentially absent even at $\tau = 0.25$, apart from one 1/300 event for workflow C at $\kappa_{\text{ref}} = 5, C = 1$.

474 3.7 Independent SEP implementation transfer

475 The independently implemented SEP endpoint produced finite measurements
476 with zero nonzero SEP flags in all 900 production rows. Its cell-resolved
477 median absolute error increased with contamination and decreased with ref-
478 erence contrast wherever contamination produced non-negligible error. At
479 $\kappa_{\text{ref}} = 1$, median absolute error was at numerical floor for $C = 0$, 0.170 for
480 $C = 0.6$, and 0.283 for $C = 1$. The corresponding contaminated-cell medians
481 were 0.0340 and 0.0566 at $\kappa_{\text{ref}} = 5$, and 0.00849 and 0.0141 at $\kappa_{\text{ref}} = 20$.

482 All six paired contamination-versus-clean contrasts had 95% sample-identity
483 bootstrap intervals strictly above zero. For the reference-contrast axis, all
484 four contaminated-domain comparisons (1 to 5 and 5 to 20 at $C = 0.6$ and
485 $C = 1$) had intervals strictly below zero for $\Delta|e| = |e|_{\kappa_{\text{high}}} - |e|_{\kappa_{\text{low}}}$. The
486 two clean-domain reference-contrast comparisons were at numerical floor and
487 were not counted as directional evidence. At the operational $\tau = 0.5$ thresh-
488 old, the observed vulnerable fraction was 0.01 for $\kappa_{\text{ref}} = 1, C = 0.6$ and 0.10
489 for $\kappa_{\text{ref}} = 1, C = 1$; it was zero in the other seven tested cells. Complete SEP
490 cell summaries and paired contrasts are reported in Tables S4–S6.

491 These results satisfy the pre-outcome transfer criterion: an independent
492 measurement implementation recovers a non-degenerate, cell-resolved stress

493 response with bootstrap-supported changes along both tested stress axes in
494 the contaminated domain. They do not establish observational validity or
495 general superiority of SEP.

496 **4 Discussion**

497 **4.1 Reliability is a landscape rather than a single ag-** 498 **gregate score**

499 The frozen AMPV V2 results are most naturally interpreted as a reliability
500 landscape over reference contrast and contamination level, rather than as a
501 single global ordering of workflows. The pooled summaries reported above
502 remain useful for orientation, but they do not identify the stress cells in
503 which workflow-relative behavior changes. The scientific role of the reliability
504 map is therefore to preserve where a performance statement holds, where it
505 weakens, and where a separately targeted validation would be required.

506 Across the tested reference-contrast grid, median absolute error and vul-
507 nerability generally decline as reference contrast increases for all workflows
508 and contamination levels. The strongest residual vulnerability occurs for
509 workflow C at low contrast, while workflow E reaches low vulnerability at

510 lower tested contrast levels. The complete workflow-by-contamination mono-
511 tonic trend listing is provided in Supplementary Note S1.

512 This pattern supports the methodological use of AMPV as a boundary-
513 mapping framework: the relevant question is not only whether two workflows
514 differ on average, but also whether their error behavior remains stable as
515 the controlled stressors move across the grid. In a pipeline-development
516 setting, such maps can identify regions that merit additional diagnostics, an
517 alternative estimator, or a separately targeted validation experiment. Those
518 downstream deployment choices are not tested here; the present contribution
519 is the evidence structure used to locate the candidate regions.

520 **4.2 Contamination changes the shape of the reliability** 521 **surface**

522 At fixed reference contrast, contamination modifies the observed error and
523 vulnerability patterns in a workflow-dependent manner. The effect is strongest
524 in the low-contrast regime and is not uniformly monotonic for every workflow-
525 contrast combination. Consequently, contamination cannot be summarized
526 independently of reference contrast or information class. The complete cell-
527 by-cell contamination trend listing is provided in Supplementary Note S2.

528 **4.3 A threshold-based view exposes operational bound-**
529 **aries**

530 Using the frozen operational vulnerability threshold $|e| > 0.5$, the lowest
531 tested reference contrast at which vulnerability is $\leq 5\%$ is summarized be-
532 low.

Workflow	C=0	C=0.6	C=1
A	5	5	5
B	5	5	5
C	20	20	20
D	5	5	5
E	1	2	2

533 This boundary table should be interpreted only within the tested grid. [6]
534 It does not imply behavior between untested reference-contrast levels, and it
535 should not be extrapolated beyond the tested contamination range.

536 4.4 Information class matters for interpretation

537 Workflow D belongs to the oracle-diagnostic information class, whereas the
538 remaining workflows are blind. Therefore, numerical differences involving D
539 should not be interpreted as if all workflows operate under the same infor-
540 mation constraints. Workflow D is used as a diagnostic reference for what
541 becomes possible with privileged information rather than as a directly inter-
542 changeable blind estimator.

543 This distinction is methodological rather than cosmetic: reliability claims
544 should remain conditional on the information available to a workflow.

545 4.5 Signed error and absolute error answer different 546 questions

547 The analysis retains both absolute and signed error. Absolute error describes
548 magnitude and is therefore used for vulnerability mapping, whereas signed er-
549 ror preserves direction. The signed-error figure is retained as supplementary
550 material so that directional structure remains inspectable without replacing
551 the primary magnitude-based reliability view.

552 A directional pattern should not be described as improved or degraded

553 performance solely from its sign. The interpretation must remain tied to
554 error magnitude, information class, and the relevant cell of the stress grid.

555 **4.6 Cross-family structure is reproducible but task-** 556 **dependent**

557 The extension does not reproduce a fixed workflow ordering, nor was that
558 the frozen transfer criterion. Across both controlled families, increasing
559 tested reference contrast is associated with lower paired absolute error wher-
560 ever nontrivial error remains, compound contamination changes the reliabil-
561 ity landscape, and matched identities resolve workflow-dependent structure.
562 However, the detailed contamination response, workflow ordering, and vul-
563 nerability boundaries differ between the comet-like and compact-PSF tasks.

564 This heterogeneity is the key methodological result rather than a failure
565 to reproduce a common ranking. In the comet-like family, contamination is
566 not uniformly monotonic and A–B behavior changes with contamination. In
567 the compact-PSF family, all tested contamination contrasts are adverse and
568 the Gaussian-fitting workflow benefits from direct model match. A pooled
569 benchmark can summarize these families but cannot by itself preserve where
570 the workflow-relative conclusions change. AMPV makes that conditionality

an explicit validation output. It should therefore be interpreted as a framework for estimating task-conditional reliability structure, not for assigning a task-independent score or universal workflow ranking.

4.7 What the present experiment does and does not establish

The study contains two single-fidelity controlled experiments. The primary comet-like experiment establishes the tested reliability landscape for its five reference-contrast levels, three contamination levels, and workflows A–E. The separately versioned compact-PSF extension establishes the corresponding landscape for blind workflows A, B, C, and E under the same stress-grid structure. The two experiments remain analytically distinct.

It does **not** by itself establish fidelity dependence across multiple simulation fidelities. [5] Any manuscript claim about fidelity dependence would require additional, explicitly authorized multi-fidelity experiments or must be framed as future work rather than as a result of the present production.

Likewise, the synthetic environment provides controlled access to ground truth and reproducible perturbations, but behavior on observational datasets outside the tested validation domain still requires external validation [5, 6].

589 The value of the framework is therefore methodological: it provides a repro-
590 ducible way to expose where a workflow remains stable, where it becomes
591 vulnerable, and which controlled stressors are associated with those changes.

592 **4.8 Limitations and next validation steps**

593 The present results are bounded by the tested grids, the two controlled syn-
594 thetic task families, the current fidelity level, and their synthetic forward-
595 model assumptions. The current analysis also uses a frozen operational vul-
596 nerability threshold of 0.5; alternative thresholds may change the location of
597 threshold-defined boundaries even when the underlying continuous error sur-
598 face is unchanged. The explicit sensitivity analysis confirms this dependence
599 and therefore supports interpreting the threshold table as an operational
600 diagnostic rather than a physical transition.

601 The SEP transfer experiment addresses one part of the previous implementation-
602 dependence limitation by showing that the matched stress-grid design can
603 be applied to a measurement endpoint outside the A–E workflow implemen-
604 tation. It does not address observational realism: the scenes remain syn-
605 thetic, the source position is known, and the transfer grid contains only three
606 tested reference-contrast levels. Useful next validation steps therefore include

607 testing additional fidelity levels, expanding the stress grid where transitions
608 appear sharp, and evaluating the framework with observationally anchored
609 injections or real-image test sets. [6] [5] These extensions should be evalu-
610 ated as new, separately versioned experiments rather than retrofitted into
611 the present analysis.

612 Interpretation of the numerical results is restricted to the tested grids,
613 information classes, and single-fidelity experiments described here.

614 5 Conclusions

615 The primary AMPV V2 comet-like experiment demonstrates a reproducible
616 way to evaluate photometric workflows as a reliability landscape rather than
617 as a single aggregate ranking. Across 300 base stochastic identities, 4,500
618 condition-specific scene instances, and 22,500 workflow evaluations, that pri-
619 mary analysis retains cell-resolved error magnitude, signed error, vulnera-
620 bility, workflow information class, and matched-scene provenance across the
621 tested reference-contrast and contamination grid.

622 The results support boundary-oriented reporting: workflow behavior varies
623 across the tested stress grid, and pooled summaries alone do not preserve the

624 full structure of those changes. The common-random-number design follows
625 an established simulation-comparison strategy [7], while the matched design
626 and versioned analysis outputs provide reproducibility and traceability in the
627 present study.

628 The compact-PSF extension, specified and frozen before outcome gen-
629 eration, broadens the evidence from one to two materially distinct con-
630 trolled synthetic photometric task families. Within these two tested fam-
631 ilies, both recover cell-resolved stress-dependent reliability structure, while
632 their workflow-relative behavior and operational boundaries differ. A fur-
633 ther pre-outcome-frozen transfer test replaced the A–E measurement end-
634 point with SEP photometry for 100 matched compact-PSF identities. All six
635 contamination-versus-clean contrasts were interval-supported as adverse, and
636 all four contaminated-domain reference-contrast comparisons were interval-
637 supported as improving with higher tested contrast. Thus the experimental-
638 design logic also produced a resolved reliability surface outside the internal
639 A–E measurement implementation, without establishing universality beyond
640 the tested synthetic validation domains.

641 The present evidence is intentionally bounded. It is a single-fidelity syn-
642 thetic study and therefore does not establish multi-fidelity dependence. More

643 generally, the synthetic evidence should not be assumed to generalize un-
644 changed to observational datasets outside the tested validation domain [5, 6].
645 Those questions require separately versioned validation experiments rather
646 than reinterpretation of the present results.

647 Within that scope, AMPV does not introduce the general V&V principle
648 that validation is domain-bounded. Its methodological contribution in this
649 astronomical workflow setting is to preserve, within a reproducible compara-
650 tive experiment, the structure that pooled benchmarking discards: matched
651 stochastic identities, explicit stress coordinates, workflow information classes,
652 continuous error surfaces, and operational vulnerability boundaries. The two
653 tested task families show why that distinction matters: broad stress-response
654 structure can be recovered reproducibly without implying a task-independent
655 workflow ordering. AMPV therefore supports validation claims about where
656 and under which tested conditions a workflow is reliable, rather than only
657 which workflow has the best pooled score.

658 **Disclosures**

659 The author declares no competing interests.

660 **Code, Data, and Materials Availability**

661 The AMPV V2 code, frozen computational outputs, numerical ledgers, veri-
662 fication records, figures, derived analyses, and supplementary reproducibility
663 materials are available at https://github.com/chenzhilei-lab/AMPV_v2.
664 No separate physical materials are associated with this computational study.

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667 **Author Contributions**

668 Zhilei Chen: Conceptualization, Methodology, Software, Validation, Formal
669 analysis, Investigation, Visualization, Writing – original draft, Writing – re-
670 view & editing.

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